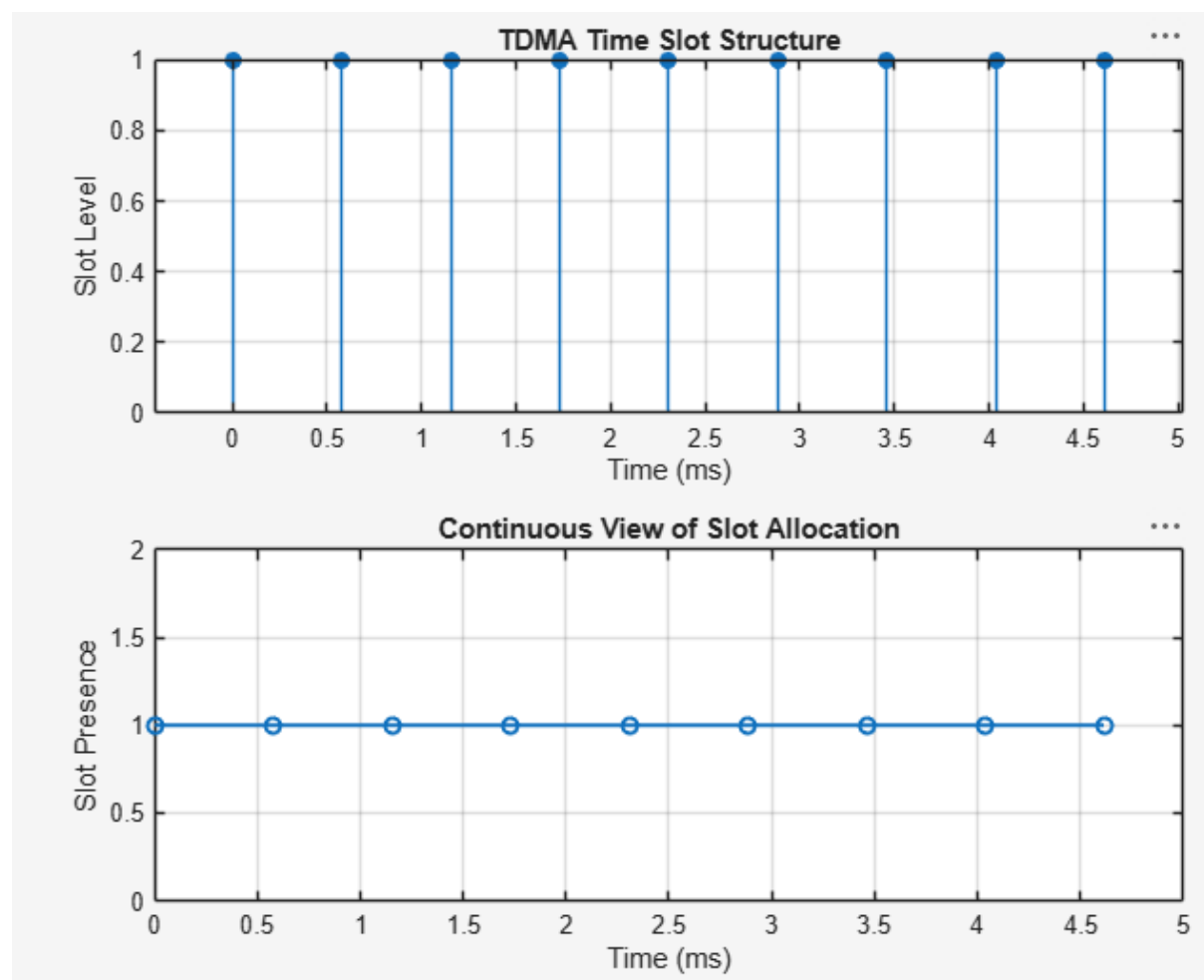


Ex.No. 2: Implementing TDMA Slot Allocation in GSM

Objective 1: Matlab Code and Output

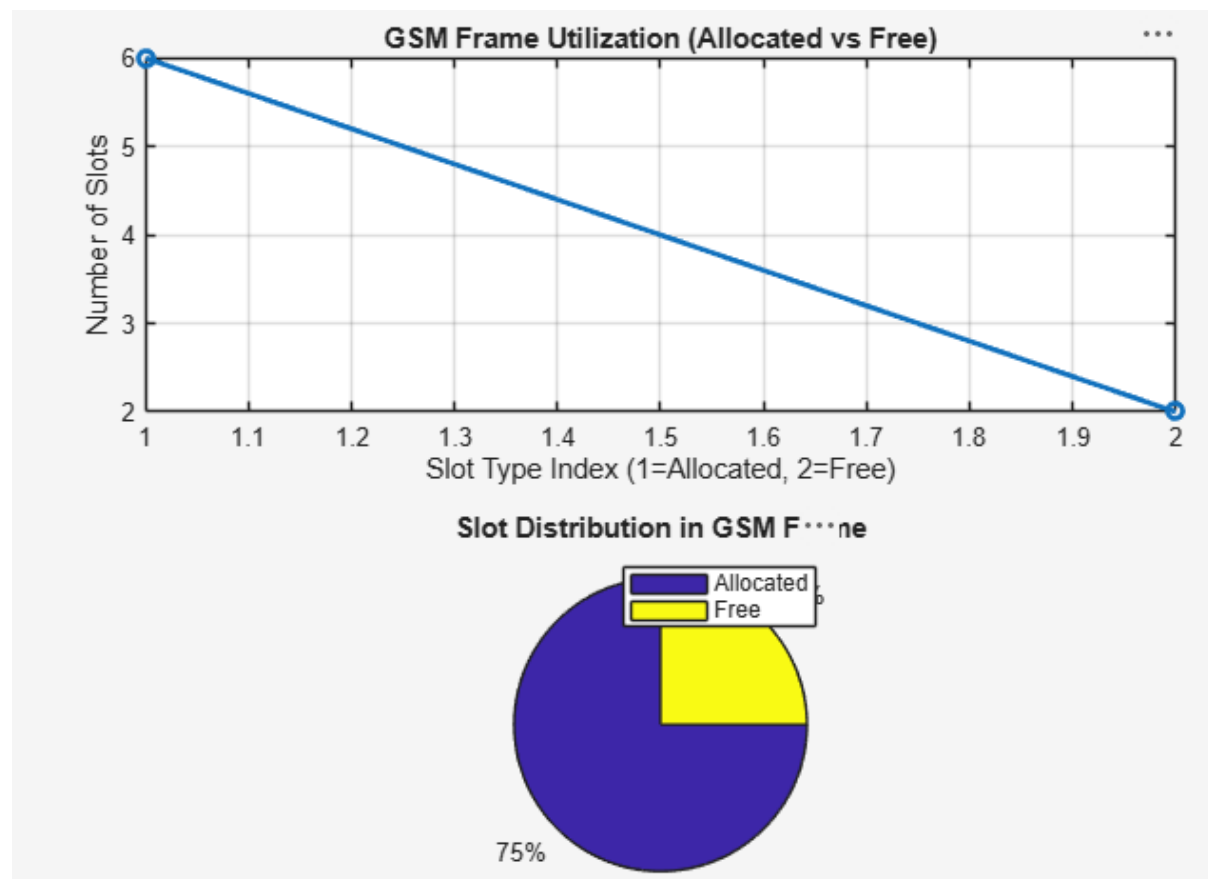
/MATLAB Drive/slotb2.m

```
1   clc; clear; close all;
2   FrameTime = 4.615; Slots = 8;
3   SlotDuration = FrameTime/Slots;
4   t = 0:SlotDuration:FrameTime;
5   figure
6   subplot(2,1,1)
7   stem(t,ones(size(t)),'filled')
8   title('TDMA Time Slot Structure')
9   xlabel('Time (ms)')
10  ylabel('Slot Level')
11  grid on
12  subplot(2,1,2)
13  plot(t,ones(size(t)),'-o','LineWidth',1.5)
14  title('Continuous View of Slot Allocation')
15
16  xlabel('Time (ms)')
17  ylabel('Slot Presence')
18  grid on
```



Objective 2: Matlab Code and Output

```
TotalSlots = 8;  
AllocatedSlots = 6;  
FreeSlots = TotalSlots - AllocatedSlots;  
Utilization = (AllocatedSlots/TotalSlots)*100;  
disp('Frame Utilization (%)')  
disp(Utilization)  
figure  
% Graph 1: Simple Line Plot (SAFE - no bar function)  
subplot(2,1,1)  
plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2)  
title('GSM Frame Utilization (Allocated vs Free)')  
xlabel('Slot Type Index (1=Allocated, 2=Free)')  
ylabel('Number of Slots')  
grid on  
  
% Graph 2: Pie Chart (SAFE)  
subplot(2,1,2)  
pie([AllocatedSlots FreeSlots])  
title('Slot Distribution in GSM Frame')  
legend('Allocated','Free')
```



Objective 3: Matlab Code and Output

```
Users = 1:8; % Total users
TotalSlots = 5; % Available slots
Alloc = zeros(1,8); % Initialize allocation
Alloc(1:TotalSlots) = 1; % Assign slots to first users
Blocked = 1 - Alloc; % Remaining users blocked
disp('User Wise Slot Allocation')
disp(table(Users',Alloc','VariableNames',{'User','Status(1=Allocated,0=Blocked)'}))
figure
% Graph 1: Allocation per user
subplot(2,1,1)

stem(Users,Alloc,'filled','LineWidth',2)
title('TDMA Slot Allocation per User')
xlabel('User Index')
ylabel('Slot Status (1=Allocated, 0=Blocked)')
ylim([-0.2 1.2])
grid on
% Graph 2: System view (safe alternative to bar)
subplot(2,1,2)
plot(Users,Alloc,'-o','LineWidth',2)
hold on
plot(Users,Blocked,'-s','LineWidth',2)
title('TDMA Slot Utilization Analysis')
xlabel('User Index')
ylabel('Status')
legend('Allocated','Blocked')
grid on
```

